1 Proposed project plan

Our project extends RESOLVE-IPD from survival outcome reconstruction to covariate-augmented downstream IPD meta-analysis. The main goal is to reconstruct plausible patient-level covariates and quantify how uncertainty in these reconstructed covariates affects downstream treatment-effect analysis.

1.1 Overall goal

RESOLVE-IPD provides reconstructed survival IPD:

$$D_j = \{(T_{ij}, \Delta_{ij}, Z_{ij})\}_{i=1}^{n_j},$$

where T_{ij} is survival time, Δ_{ij} is the event indicator, and Z_{ij} is treatment assignment.

Our goal is to augment this dataset with reconstructed patient-level covariates:

$$X_{ij} = (X_{ij1}, X_{ij2}, \ldots, X_{ijp}),$$

such as age group, sex, ECOG status, region, and PD-L1 status.

Because published studies usually report only marginal covariate summaries, the joint covariate distribution is not identifiable. Therefore, instead of constructing one imputed dataset, we construct a set of plausible joint covariate reconstructions:

$$\mathcal{X}_j = \{X_j^{(1)}, X_j^{(2)}, \ldots, X_j^{(B)}\}.$$

1.2 Step 1: reconstruct marginal covariates

First, for each study j and covariate k, we extract published marginal summary information from baseline tables or subgroup reports. For example, if the paper reports the number of patients in category a, we impose

$$\sum_{i=1}^{n_j} \mathbf{1}(X_{ijk} = a) = n_{jk,a}.$$

This step ensures that each reconstructed covariate matches the reported marginal distribution.

1.3 Step 2: reconstruct joint covariate labels

The second step is to reconstruct patient-level joint covariate labels:

$$X_{ij} = (X_{ij1}, X_{ij2}, \ldots, X_{ijp}).$$

We will construct a feasible set:

$$\mathcal{X}_j = \{X_j : X_j \text{ matches all available published constraints}\}.$$

At minimum, X_j must satisfy marginal constraints:

$$\sum_{i=1}^{n_j} \mathbf{1}(X_{ijk} = a) = n_{jk,a}.$$

If additional joint information is available, we also include pairwise constraints:

$$\sum_{i=1}^{n_j} \mathbf{1}(X_{ijk} = a, X_{ij\ell} = b) = n_{jk\ell,ab}.$$

If exact joint counts are unavailable, we can use external or clinical information as soft constraints:

$$\mathrm{Corr}(X_{jk}, X_{j\ell}) \approx \rho_{k\ell}^{\mathrm{ext}}.$$

The output of this step is a collection of plausible reconstructions:

$$\mathcal{X}_j = \{X_j^{(1)}, X_j^{(2)}, \ldots, X_j^{(B)}\}.$$

1.4 Step 3: propagate covariate reconstruction uncertainty

For each plausible reconstruction $X^{(b)}$, we repeat the same downstream analysis:

$$Q^{(b)} = A(D, X^{(b)}), \qquad b = 1, \ldots, B,$$

where $A(\cdot)$ is the downstream IPD meta-analysis procedure and $Q^{(b)}$ is the resulting estimate.

The final result is a distribution:

$$\{Q^{(1)}, Q^{(2)}, \ldots, Q^{(B)}\}.$$

This distribution measures how sensitive the downstream conclusion is to uncertainty in covariate reconstruction.

1.5 Step 4: evaluate adjusted treatment effects

The first downstream analysis is covariate-adjusted treatment effect estimation. For each reconstructed covariate matrix $X^{(b)}$, we fit

$$h_{ij}(t) = h_{0j}(t) \exp\left\{\alpha Z_{ij} + \beta^\top X_{ij}^{(b)}\right\}.$$

We then compare the resulting treatment effect estimates:

$$\widehat{\alpha}^{(1)}, \widehat{\alpha}^{(2)}, \ldots, \widehat{\alpha}^{(B)}.$$

1.6 Step 5: evaluate treatment-effect heterogeneity

The second downstream analysis is treatment-effect heterogeneity. We fit

$$h_{ij}(t) = h_{0j}(t) \exp\left\{\alpha Z_{ij} + \beta^\top X_{ij}^{(b)} + \gamma^\top (Z_{ij} X_{ij}^{(b)})\right\}.$$

Here, γ_k measures whether covariate k modifies the treatment effect.

If many covariates are included, we use shrinkage on the interaction terms:

$$\min_{\alpha,\beta,\gamma} \left\{-\ell(\alpha, \beta, \gamma) + \lambda \sum_{k=1}^{p} |\gamma_k|\right\}.$$

For each covariate k, we measure how often it is selected as an effect modifier:

$$\widehat{\pi}_k = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\left(\widehat{\gamma}_k^{(b)} \neq 0\right).$$